



# Topics

- What is Jenkins and CI/CD?
- Lab Setup & Installation
- Jenkins Dashboard Overview
- Some Basic Jenkins Job
- Scheduling The Job
- Working with GIT Repo
- Email Notification
- Job for action on Remote server
- Jenkins + Ansible Playbook
- Project - Website Update on Remote Server
- User Management (Role Based)
- Environment Variables
- Jenkins + MAVEN
- CI/CD Project (Jenkins PIPELINE)
- Bonus Tips



Jenkins

# What is Jenkins?





**Jenkins is an open-source  
automation server**



**Automate various parts of  
software development.**

**Building**

**Testing**

**Deploy**





### Continuous Integration and Continuous Delivery

As an extensible automation server, Jenkins can be used as a simple CI server or turned into the continuous delivery hub for any project.



### Easy installation

Jenkins is a self-contained Java-based program, ready to run out-of-the-box, with packages for Windows, Linux, macOS and other Unix-like operating systems.



### Easy configuration

Jenkins can be easily set up and configured via its web interface, which includes on-the-fly error checks and built-in help.



### Plugins

With hundreds of plugins in the Update Center, Jenkins integrates with practically every tool in the continuous integration and continuous delivery toolchain.



### Extensible

Jenkins can be extended via its plugin architecture, providing nearly infinite possibilities for what Jenkins can do.



### Distributed

Jenkins can easily distribute work across multiple machines, helping drive builds, tests and deployments across multiple platforms faster.



# What is CI/CD?

**CI/CD is a method of frequently delivering apps to customers by introducing automation into the stages of app development.**

## CONTINUOUS INTEGRATION



## CONTINUOUS DELIVERY



## CONTINUOUS DEPLOYMENT





# Lab Setup

# Installation



```
sudo dnf upgrade
```

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \  
https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

```
sudo rpm --import https://pkg.jenkins.io/redhat-  
stable/jenkins.io-2023.key
```

```
sudo yum install jenkins
```

```
sudo systemctl start/stop jenkins
```

To start the Jenkins from browser: IP:8080

**To access it from browser, we need to enable it on firewall**

---

```
sudo firewall-cmd --permanent --zone=public --add-port=8080/tcp  
sudo firewall-cmd --reload
```



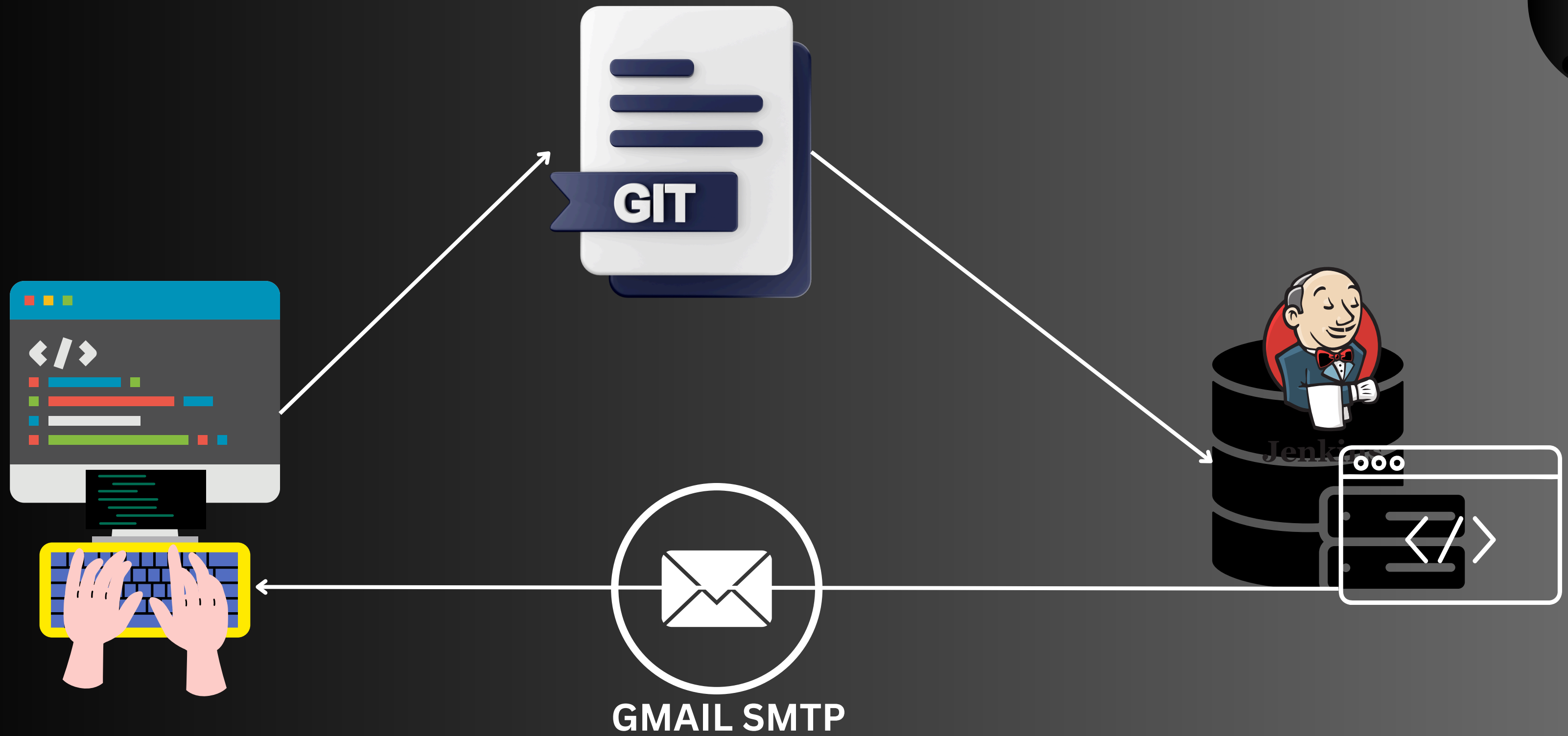
Jenkins

# Overview Of DASHBOARD



[Git hub link - https://github.com/paulphilip/pythoncode](https://github.com/paulphilip/pythoncode)





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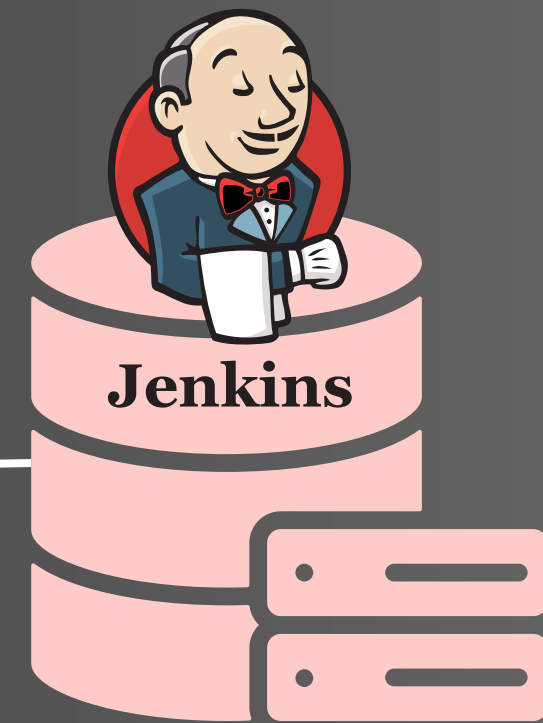


Task like

Transfer file  
Execute Commands



SERVER B



SERVER A



Jenkins





ServerA



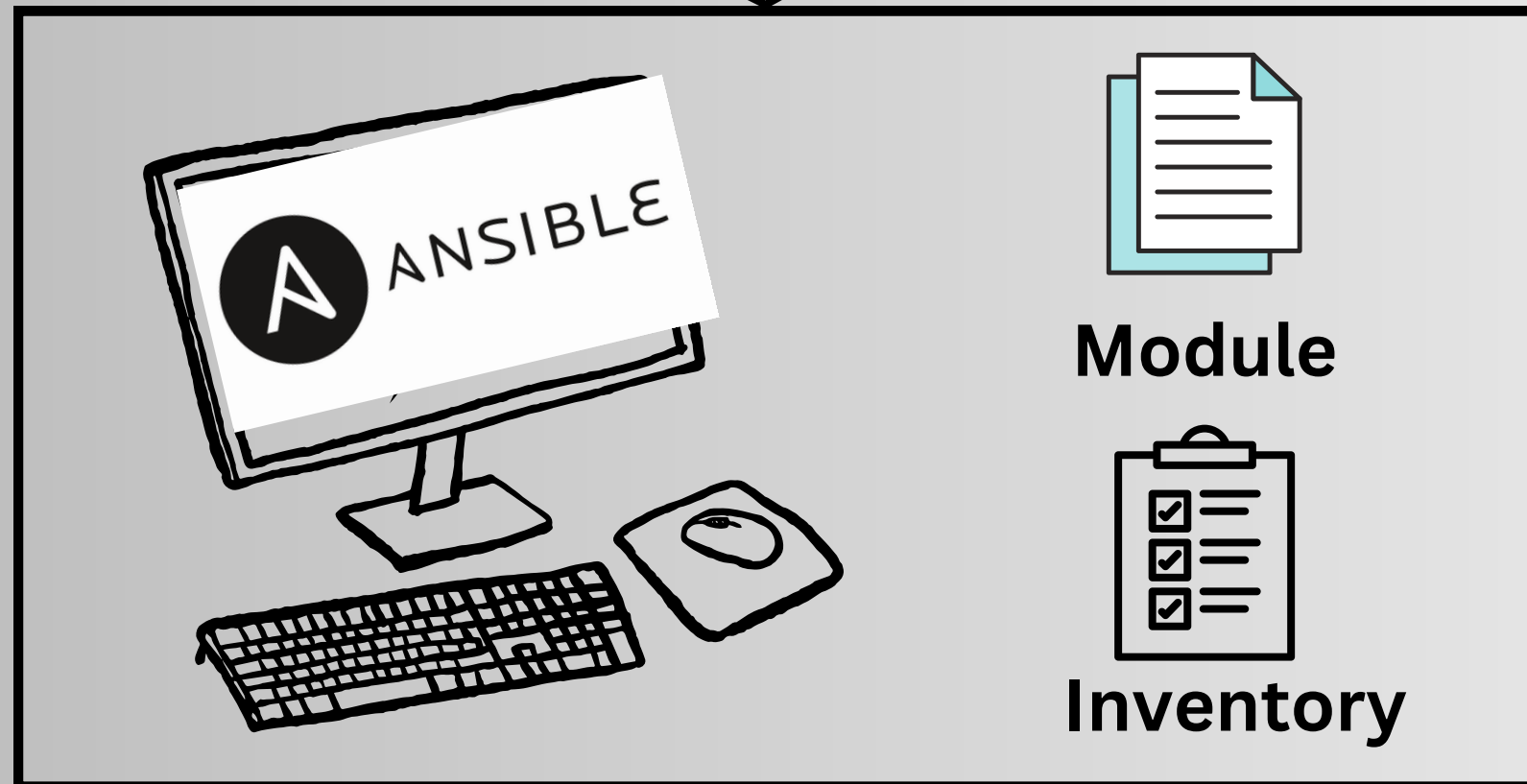
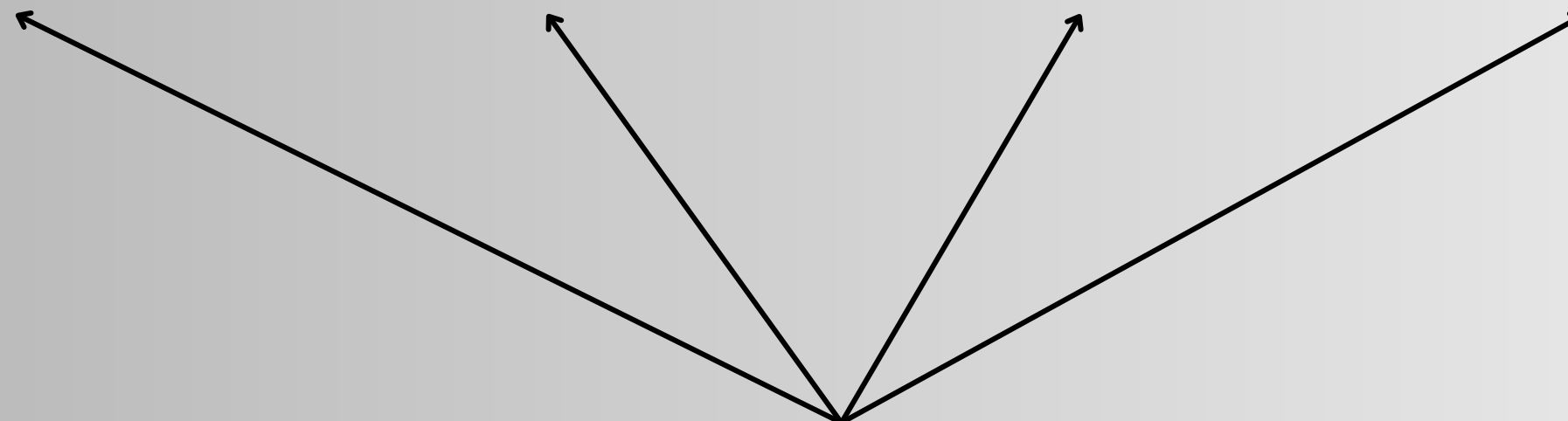
ServerB



ServerC



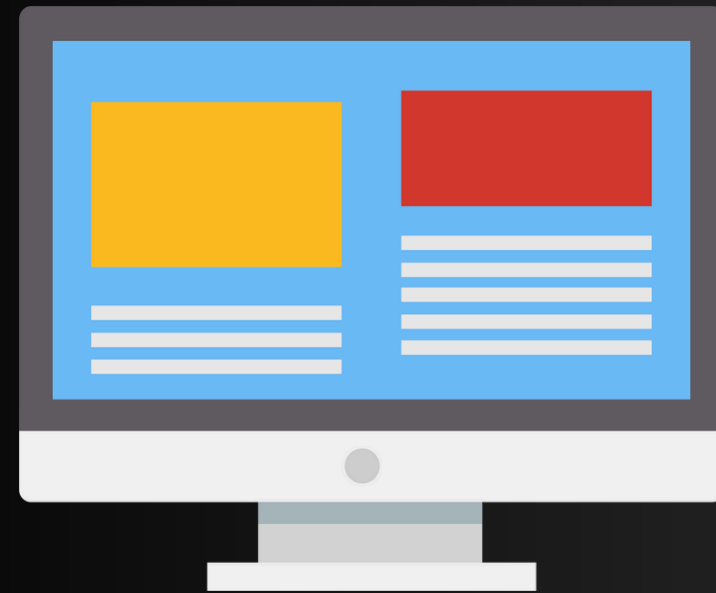
ServerD





Jenkins

SERVER A



Apache Webserver



SERVER B

# Summary of Jenkins + Ansible Section

- Quick overview of Ansible
- Installing Ansible plugin
- Jenkins job with 'Invoke Ansible Playbook'
- Perform task on Local vs Remote host
- Ansible job execute as 'jenkins' user
- Execute task which required SUDO access
- Basic project of Updating the Webpage of remote server and publish it.

# User Management

- Plugin: Role-based Authorization Strategy
- Create a new user

## Manage Jenkins:

- Security -> Authorization -> Role-based
- Manage and Assign Roles
  - Add a new read-only role and assign the permission
  - Assign Roles -> Add new user (nick) and assign read-only role

# Environment Variables

We can use pre-defined variables by jenkins like

**BUILD\_NUMBER, WORKSPACE etc**

[https://wiki.jenkins.io/display/JENKINS/Building+a+software+project#Buildingasoftwar  
eproject-belowJenkinsSetEnvironmentVariables](https://wiki.jenkins.io/display/JENKINS/Building+a+software+project#Buildingasoftwarproject-belowJenkinsSetEnvironmentVariables)





Jenkins

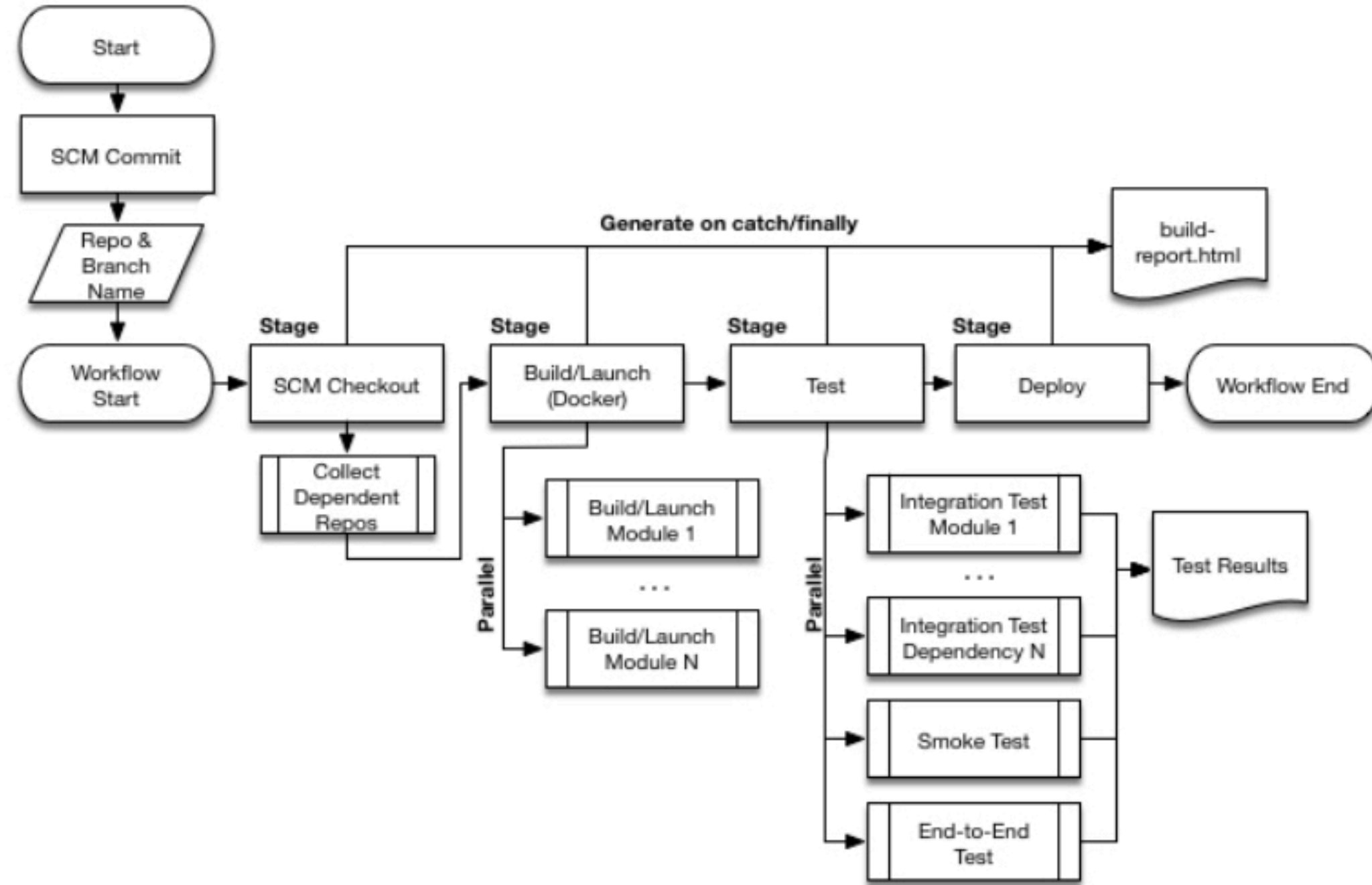
# What is maven?

**Maven is a build automation tool used primarily for Java projects.**

- **Working with Git and Maven**
- **Build jar using Maven**
- **Testing**
- **Deploy jar locally**
- **Graphical representation of results**
- **Send Email notification**



# Jenkins Pipeline



<https://www.jenkins.io/doc/book/pipeline/>